

The Robinson annulation is a three-step reaction that forms a six-membered ring containing an α,β -unsaturated ketone. The reaction begins with a Michael addition, which involves an acid- or base-catalyzed conjugate addition of an enolate nucleophile (made from Compound **1**) to a vinyl ketone (Compound **2**), forming Compound **3**. The Michael addition is followed by an intramolecular aldol reaction, in which Compound **3** cyclizes to form a β -hydroxy ketone (Compound **4**). Finally, dehydration occurs and yields an α,β -unsaturated ketone (Compound **5**). The three reactions occurring in the Robinson annulation are shown in figures 1–3.

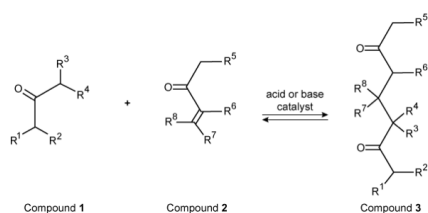


Figure 1 Michael addition (Note: R^1 through R^8 = H, alkyl, or aryl)

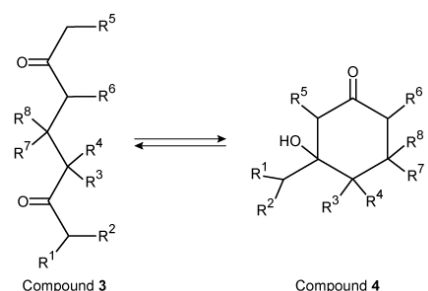


Figure 2 Intramolecular aldol reaction (Note: R^1 through R^8 = H, alkyl, or aryl)

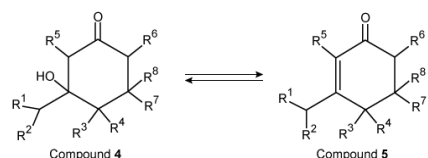


Figure 3 Dehydration (Note: R^1 through R^8 = H, alkyl, or aryl)

The Hajos-Parrish reaction is a special case of the Robinson annulation in which the intramolecular aldol reaction is asymmetric and catalyzed by either of the chiral amino acids shown in Figure 4.

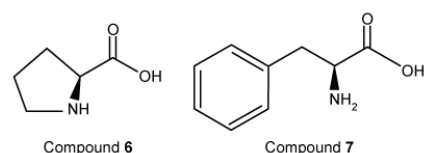


Figure 4 Structure of chiral amino acid catalysts

The Hajos-Parrish reaction has been applied in the syntheses of many organic molecules including (+)-desogestrel (Figure 5), which is a progesterone receptor agonist.

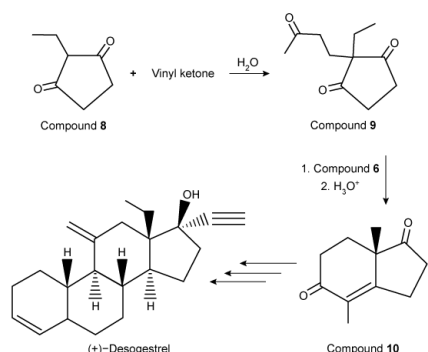


Figure 5 Hajos-Parrish reaction used in the synthesis of (+)-desogestrel

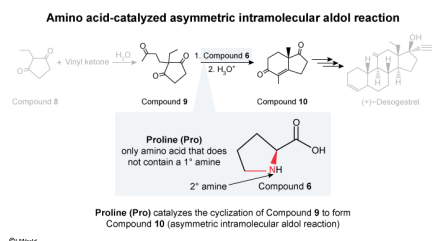
Which of the following amino acids is used to catalyze the intramolecular aldol step of the Hajos-Parrish reaction in Figure 5?

- ☐ A. Tyr [12%]
☐ B. Phe [11%]
☐ C. Trp [13%]
☒ D. Pro [62%]

Correct

62% answered correctly

Explanation:



Amino acids are important biological molecules that make up proteins. The **structure** of an amino acid consists of an **amino group**, a **carboxyl group**, and a **side chain**, all attached to a central carbon atom (ie, the **α -carbon**). Amino acids differ only in the structure of the side chain and based on the nature of that side chain, amino acids can be **classified** as **hydrophobic** (nonpolar) or **hydrophilic** (polar). The polar amino acids can be further classified as **acidic** (ie, negatively charged at physiological pH), **basic** (ie, positively charged at physiological pH), or **uncharged**. In addition to their chemical names, amino acids have one- and three-letter abbreviations.

Proline (Pro, P) is a nonpolar amino acid containing a **pyrrolidine** side chain. In proline, the nitrogen atom is that of both the pyrrolidine ring and the amino group in the amino acid backbone. Proline is the only amino acid with a **secondary amine** in its backbone. The chiral amino acid (ie, **Compound 6**) used in the Hajos-Parrish reaction in Figure 5 contains a pyrrolidine side chain. Therefore, **Compound 6** is the amino acid proline (Pro).

(Choice A) **Tyrosine** (Tyr, Y) is a polar amino acid containing a phenol side chain.

(Choice B) **Compound 7** is **phenylalanine** (Phe, F), which is a nonpolar amino acid containing a phenyl ring side chain. Although this is one of the amino acid catalysts given in the passage, Figure 5 shows that **Compound 6** was used in the Hajos-Parrish reaction to form (+)-desogestrel.

(Choice C) **Tryptophan** (Trp, W) is a nonpolar amino acid containing an indole side chain. Although Trp does have a secondary amine like Pro, the secondary amine is part of the side chain and is not the amino group in the amino acid backbone.

Educational objective:

Amino acids consist of an amino group, a carboxyl group, and a side chain, all bonded to a central carbon atom (ie, the α -carbon). Amino acids are classified in several ways based on the side chain functional group. Proline (Pro, P) is a nonpolar amino acid containing a pyrrolidine side chain; it is the only amino acid in which the backbone amino group is a secondary amine.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 403369
System:
Functional Groups and Their Reactions

The Robinson annulation is a three-step reaction that forms a six-membered ring containing an α,β -unsaturated ketone. The reaction begins with a Michael addition, which involves an acid- or base-catalyzed conjugate addition of an enolate nucleophile (made from Compound **1**) to a vinyl ketone (Compound **2**), forming Compound **3**. The Michael addition is followed by an intramolecular aldol reaction, in which Compound **3** cyclizes to form a β -hydroxy ketone (Compound **4**). Finally, dehydration occurs and yields an α,β -unsaturated ketone (Compound **5**). The three reactions occurring in the Robinson annulation are shown in figures 1–3.

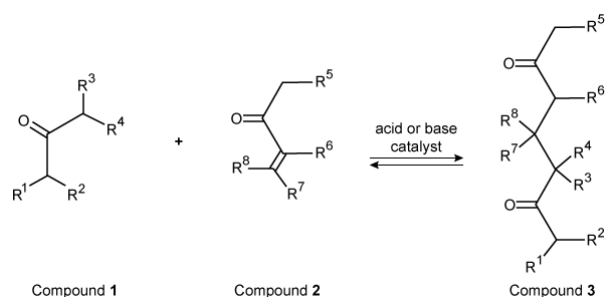


Figure 1 Michael addition (Note: R^1 through R^8 = H, alkyl, or aryl)

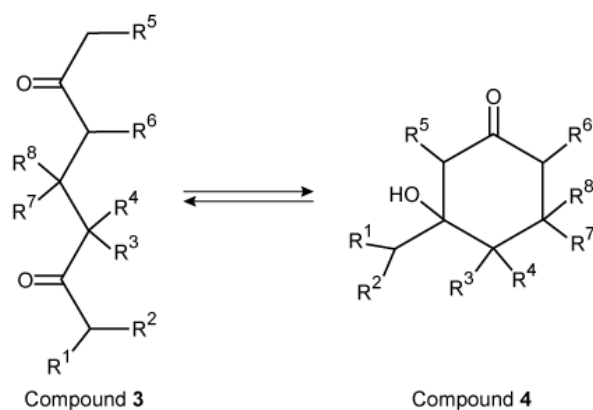


Figure 2 Intramolecular aldol reaction (Note: R^1 through R^8 = H, alkyl, or aryl)

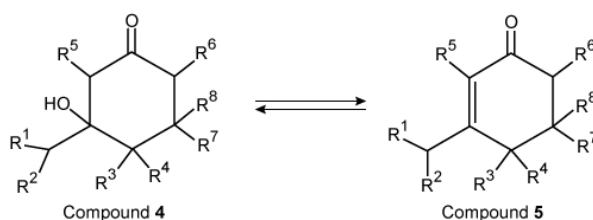


Figure 3 Dehydration (Note: R^1 through R^8 = H, alkyl, or aryl)

The Hajos-Parrish reaction is a special case of the Robinson annulation in which the intramolecular aldol reaction is asymmetric and catalyzed by either of the chiral amino acids shown in Figure 4.

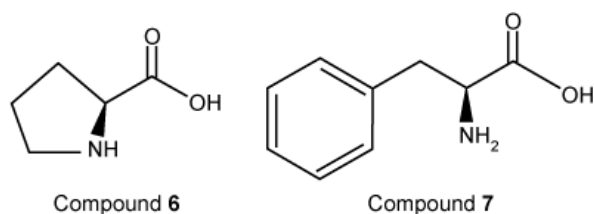


Figure 4 Structure of chiral amino acid catalysts

The Hajos-Parrish reaction has been applied in the syntheses of many organic molecules including (+)-desogestrel (Figure 5), which is a progesterone receptor agonist.

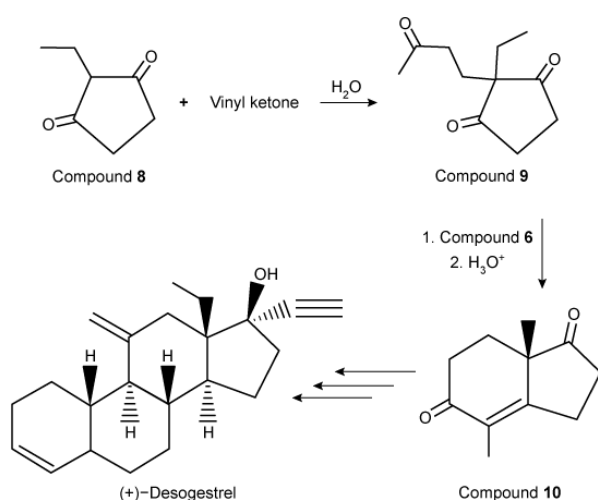


Figure 5 Hajos-Parrish reaction used in the synthesis of (+)-desogestrel

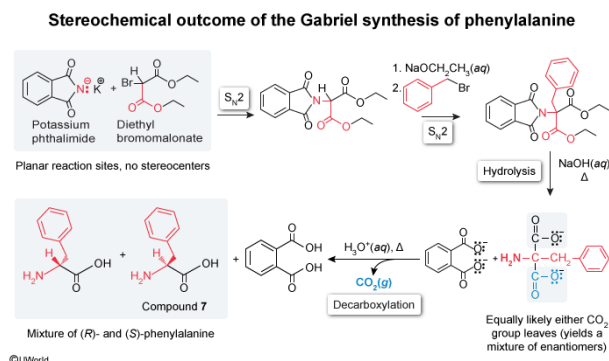
The Gabriel synthesis is not an ideal method to synthesize Compound 7 for use in the Hajos-Parrish reaction. Which of the following best explains why this is true?

- ☐ A. The Gabriel synthesis results in a mixture of diastereomers, and the catalyst used in the Hajos-Parrish reaction is a single enantiomer. [12%]
- ✓ ☒ B. The Gabriel synthesis results in a mixture of enantiomers, and the catalyst used in the Hajos-Parrish reaction is a single enantiomer. [59%]
- ☐ C. The Gabriel synthesis results in the *S* enantiomer, and the catalyst used in the Hajos-Parrish reaction is the *R* enantiomer. [10%]
- ☐ D. The Gabriel synthesis results in the *R* enantiomer, and the catalyst used in the Hajos-Parrish reaction is the *S* enantiomer. [16%]

Correct

60% answered correctly

Explanation:



The **Gabriel synthesis**, also known as the malonic ester synthesis, is used to make **primary amines** (including **α -amino acids**) without overalkylation of the amine. For the synthesis of amino acids, potassium phthalimide (ie, a "protected" form of ammonia) acts as a nucleophile and attacks the electrophilic carbon of diethyl bromomalonate in an **S_N2 reaction**, where bromide is the **leaving group**. In a subsequent S_N2 reaction, the **α -carbon** is deprotonated with a base and attacks an alkyl halide to add the amino acid side chain. **Hydrolysis** of the phthalimide group and the esters on the α -carbon is followed by acidification of the carboxyl groups and **decarboxylation**, yielding an amino acid.

The amino acid synthesis begins with no chiral centers in place because **potassium phthalimide** and **diethyl bromomalonate** have **planar** C=O groups and do not have **chiral centers**. The **decarboxylation step** creates a stereocenter at the α -carbon of the amino acid, but the reaction is **not stereospecific** because an equal probability exists that either carboxyl group could leave as CO₂. Therefore, the Gabriel synthesis results in a mixture of **enantiomers** (ie, a mixture of (*R*)- and (*S*)-amino acids).

Based on Compound **7** in Figure 4, the catalyst used in the asymmetric intramolecular aldol step of the Hajos-Parrish reaction is a single enantiomer. As such, the Gabriel synthesis is not an ideal method to make Compound **7** because the synthesis results in a mixture of enantiomers.

(Choice A) The Gabriel synthesis results in a mixture of *enantiomers*, not **diastereomers**.

(Choices C and D) Because the final step in the Gabriel synthesis is not stereospecific, the final product is *not* a single enantiomer. Based on the structure of Compound **7** in Figure 4, the chiral amino acid catalyst used in the Hajos-Parrish reaction in Figure 5 is **(*S*)-phenylalanine**.

Educational objective:

The Gabriel synthesis is a method used to make α -amino acids with potassium phthalimide and diethyl bromomalonate, which are starting materials that have planar reaction sites without chiral centers. Although the decarboxylation step creates a stereocenter, this step is not stereospecific and causes the Gabriel synthesis to yield a mixture of enantiomers.

Time spent:0 sec**Subject:**
Organic Chemistry**QID:**403370**System:**
Functional Groups and Their Reactions

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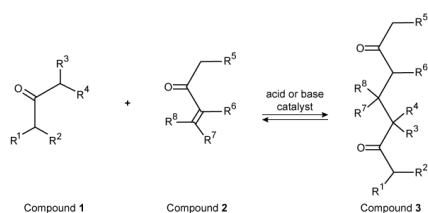


Figure 1 Michael addition (Note: R^1 through R^8 = H, alkyl, or aryl)

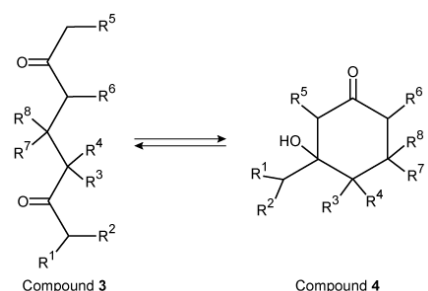


Figure 2 Intramolecular aldol reaction (Note: R^1 through R^8 = H, alkyl, or aryl)

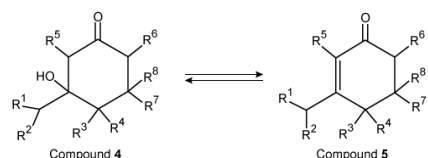


Figure 3 Dehydration (Note: R^1 through R^8 = H, alkyl, or aryl)

The Hajos-Parrish reaction is a special case of the Robinson annulation in which the intramolecular aldol reaction is asymmetric and catalyzed by either of the chiral amino acids shown in Figure 4.

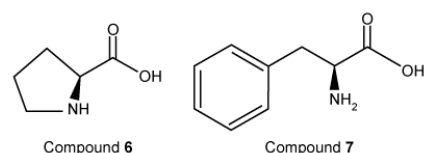


Figure 4 Structure of chiral amino acid catalysts

The Hajos-Parrish reaction has been applied in the syntheses of many organic molecules including (+)-desogestrel (Figure 5), which is a progesterone receptor agonist.

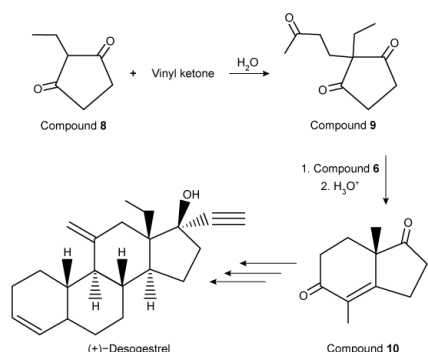
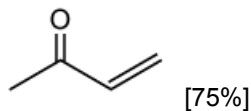


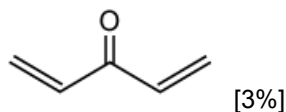
Figure 5 Hajos-Parrish reaction used in the synthesis of (+)-desogestrel

Based on the information in Figure 5, which of the following compounds is the vinyl ketone used in the synthesis of (+)-desogestrel?

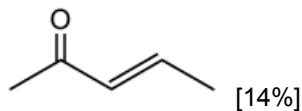
✓ ☒ A.



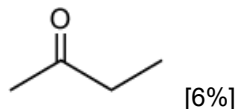
☐ B.



☐ C.



☐ D.



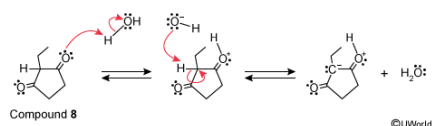
Correct

75% answered correctly

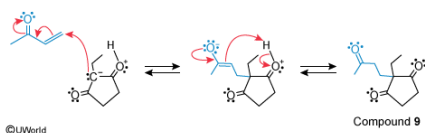
Explanation:

α,β -Unsaturated carbonyl compounds (ie, vinyl carbonyl compounds) are molecules with an **electrophilic double bond**. Through **resonance**, the partial positive charge (δ^+) is **distributed** between the carbonyl carbon (C=O) and the β -carbon.

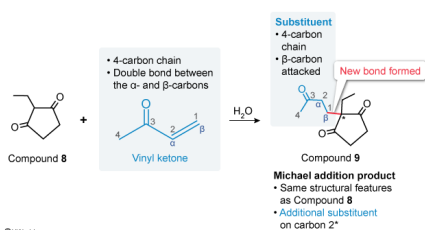
The first step in the Hajos-Parrish reaction shown in Figure 5 is a **Michael addition**, which involves a **1,4-addition** (ie, a conjugate addition) of an enolate nucleophile to the β -carbon of an α,β -unsaturated carbonyl compound. The reaction begins with enolate formation via protonation of a carbonyl on Compound **8** and then acidic hydrogen deprotonation.



Nucleophilic attack on the β -carbon of a vinyl ketone then occurs, followed by proton transfer to give the final product of the Michael addition (Compound **9**).



Because the Michael addition involves nucleophilic addition to the β -carbon of an α,β -unsaturated carbonyl compound, the vinyl ketone that reacts in Figure 5 must contain a double bond between the carbon that bonds to the five-membered ring and the adjacent carbon (ie, the α and β carbons).



Therefore, the **vinyl ketone** is a four-carbon chain with a ketone adjacent to a double bond. **Choice A** shows the only diagram that contains these structural features.

(Choice B) This structure contains too many carbons and an extra double bond. Because the Michael addition adds to *one* double bond, any additional double bonds would be **observed** in the product.

(Choice C) This structure has a methyl group bonded to the β -carbon, which would be **observed** as an additional substituent.

(Choice D) This structure is *not* a vinyl ketone because it does not contain a double bond.

Educational objective:

α,β -Unsaturated carbonyl compounds contain electrophilic double bonds. These molecules are electrophiles in the Michael addition, which involves a 1,4-addition (ie, conjugate addition) of an enolate to the β -carbon of an α,β -unsaturated carbonyl compound.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403371

System:
Functional Groups and Their Reactions

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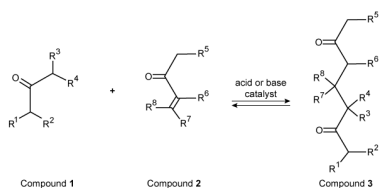


Figure 1 Michael addition (Note: R^1 through R^8 = H, alkyl, or aryl)

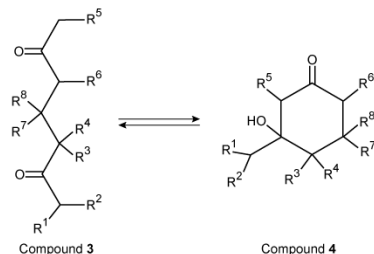


Figure 2 Intramolecular aldol reaction (Note: R^1 through R^8 = H, alkyl, or aryl)

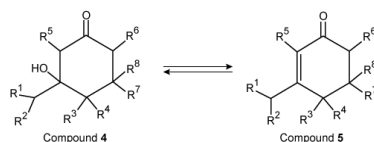


Figure 3 Dehydration (Note: R^1 through R^8 = H, alkyl, or aryl)

The Hajos-Parrish reaction is a special case of the Robinson annulation in which the intramolecular aldol reaction is asymmetric and catalyzed by either of the chiral amino acids shown in Figure 4.

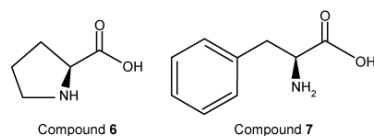


Figure 4 Structure of chiral amino acid catalysts

The Hajos-Parrish reaction has been applied in the syntheses of many organic molecules including (+)-desogestrel (Figure 5), which is a progesterone receptor agonist.

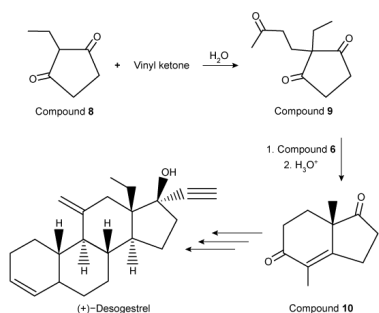


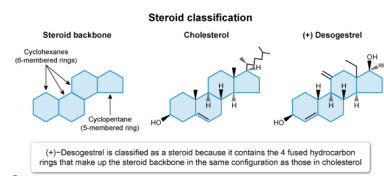
Figure 5 Hajos-Parrish reaction used in the synthesis of (+)-desogestrel

Consider the structure of (+)-desogestrel shown in Figure 5. Can this compound be classified as a steroid?

- ☐ A. Yes, because (+)-desogestrel is a triterpene. [5%]
- ☐ B. No, because (+)-desogestrel is made of isoprene units. [2%]
- ☒ C. Yes, because (+)-desogestrel contains four fused hydrocarbon rings with the same configuration as those in cholesterol. [83%]
- ☐ D. No, because (+)-desogestrel does not contain five fused hydrocarbon rings with the same configuration as those in cholesterol. [8%]

Correct
83% answered correctly

Explanation:



Terpenes are precursors to **steroids** (ie, lipids involved in signaling pathways). Individual terpenes are made up of two or more 5-carbon groups known as isoprenes, which can be joined together in one of **three ways**. Terpenes are **classified** based on the number of isoprene units present. Steroids are made up of terpenes that form a **backbone structure** containing **four fused hydrocarbon rings**: three 6-membered rings and one 5-membered ring. The steroid backbone structure is made by the **cyclization of squalene**, as in the synthesis of cholesterol.

The structure of (+)-desogestrel shown in Figure 5 contains three 6-membered rings and one 5-membered ring fused together in the same configuration as the hydrocarbon rings in cholesterol (ie, the four fused hydrocarbon rings in a steroid backbone). Therefore, (+)-desogestrel can be classified as a steroid.

(Choice A) Triterpenes are made of three monoterpenes, or **six isoprene units** (ie, 30 carbon atoms). (+)-Desogestrel is made of 22 carbon atoms, and therefore is *not* a triterpene.

(Choice B) Steroids *are* made of isoprene units, and (+)-desogestrel *is* made of isoprenes.

(Choice D) Cholesterol contains a steroid backbone made of *four* fused hydrocarbon rings, not five fused hydrocarbon rings.

Educational objective:

Steroids are made of terpenes, which are composed of two or more isoprenes (5-carbon groups). A steroid backbone contains four fused hydrocarbon rings consisting of three 6-membered rings and one 5-membered ring, with the configuration obtained from the cyclization of squalene to make cholesterol.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 403372
System:
Functional Groups and Their Reactions

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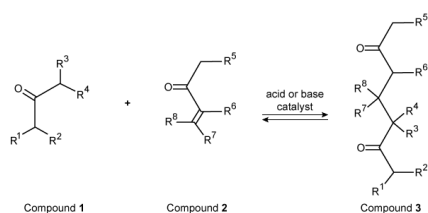


Figure 1 Michael addition (Note: R^1 through R^8 = H, alkyl, or aryl)

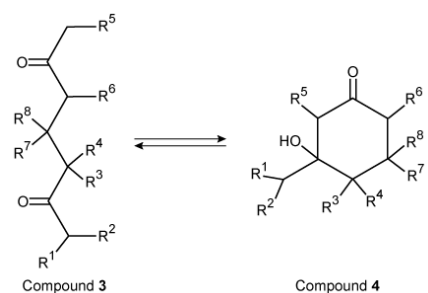


Figure 2 Intramolecular aldol reaction (Note: R^1 through R^8 = H, alkyl, or aryl)

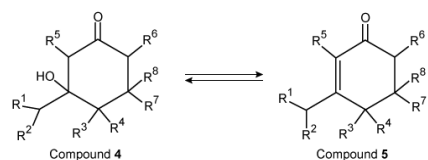


Figure 3 Dehydration (Note: R^1 through R^8 = H, alkyl, or aryl)

The Hajos-Parrish reaction is a special case of the Robinson annulation in which the intramolecular aldol reaction is asymmetric and catalyzed by either of the chiral amino acids shown in Figure 4.

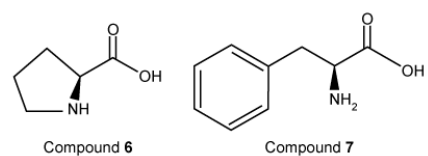


Figure 4 Structure of chiral amino acid catalysts

The Hajos-Parrish reaction has been applied in the syntheses of many organic molecules including (+)-desogestrel (Figure 5), which is a progesterone receptor agonist.

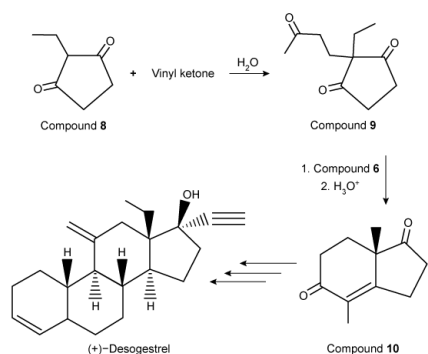


Figure 5 Hajos-Parrish reaction used in the synthesis of (+)-desogestrel

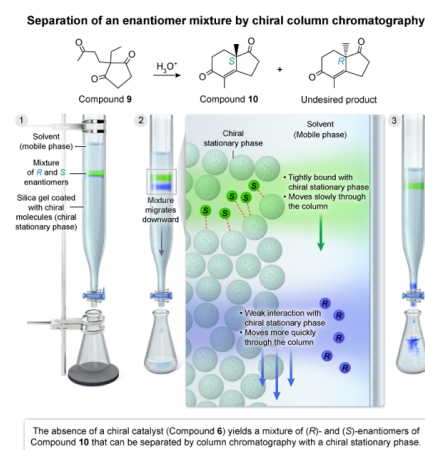
How could Compound **10** be isolated if Compound **6** were not used in the synthesis of (+)-desogestrel?

- ☐ A. Use a resolving agent to create a mixture of enantiomers, which can be separated due to having different chemical properties. [21%]
- ☐ B. Use a resolving agent to create a mixture of diastereomers, which can be separated due to having different chemical properties. [33%]
- ☐ C. Separate the mixture of diastereomers obtained from the reaction using column chromatography with a chiral stationary phase. [11%]
- ☒ D. Separate the mixture of enantiomers obtained from the reaction using column chromatography with a chiral stationary phase. [33%]

Correct

32% answered correctly

Explanation:



A carbon atom bonded to four different substituents is a **chiral center**. If two molecules have the same atom connectivity but opposite configuration at each chiral center (eg, *R* in one molecule and *S* in the other molecule), the two molecules are **enantiomers**.

Mixtures of enantiomers are not readily separated because enantiomers have the **same chemical and physical properties**. Consequently, if the physical properties of the molecules are not modified prior to separation by the addition of a **resolving agent**, the separation of the enantiomers can only be achieved by a separation technique that uses chiral media. **Chiral column chromatography** uses a chiral stationary phase, which interacts with one enantiomer (either *R* or *S*) more strongly, causing that enantiomer to move slowly through the column while the other enantiomer with a weaker interaction moves through the column more quickly.

The passage states that the asymmetric **intramolecular aldol condensation** (ie, second step in Figure 5) requires a chiral amino acid catalyst (Compound **6** in this reaction) to produce Compound **10**. Therefore, if Compound **6** is not used, the reaction will *not* be asymmetric (ie, the chiral center formed will be a mixture of *R* and *S*), and because Compound **10** contains only one stereocenter, a mixture of enantiomers will be formed. Accordingly, chiral column chromatography can be used to separate the enantiomer mixture.

(Choice A) Using a resolving agent creates a mixture of *diastereomers* rather than enantiomers.

(Choice B) Separations (eg, column chromatography, distillation) rely on differences in *physical* properties rather than chemical properties.

(Choice C) **Diastereomers** are stereoisomers with *two* or more chiral centers in which some, but not all, of the chiral centers are opposite. Because Compound **10** only has only one chiral center, the absence of a catalyst results in a mixture of enantiomers, not diastereomers.

Educational objective:

Because enantiomers have the same chemical and physical properties, they are not separable unless the molecules are modified prior to separation using a resolving agent or the compounds are passed through chiral media. Chiral column chromatography uses a chiral stationary phase that interacts more strongly with one enantiomer while the other enantiomer moves more quickly through the column.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403373
System:
Functional Groups and Their Reactions